

Complete Technology Stack

RASTA-AI — AI-Based Smart Logistics and Accessibility
Intelligence Platform for the North Eastern Region

Problem Statement	SIH26002 — AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
Organisation	Ministry of Development of North Eastern Region (MDoNER)
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Every version number, provider URL, table name and endpoint in this document was read out of the working tree on the date above.
Nothing here is aspirational unless it is explicitly tagged **PLANNED**.

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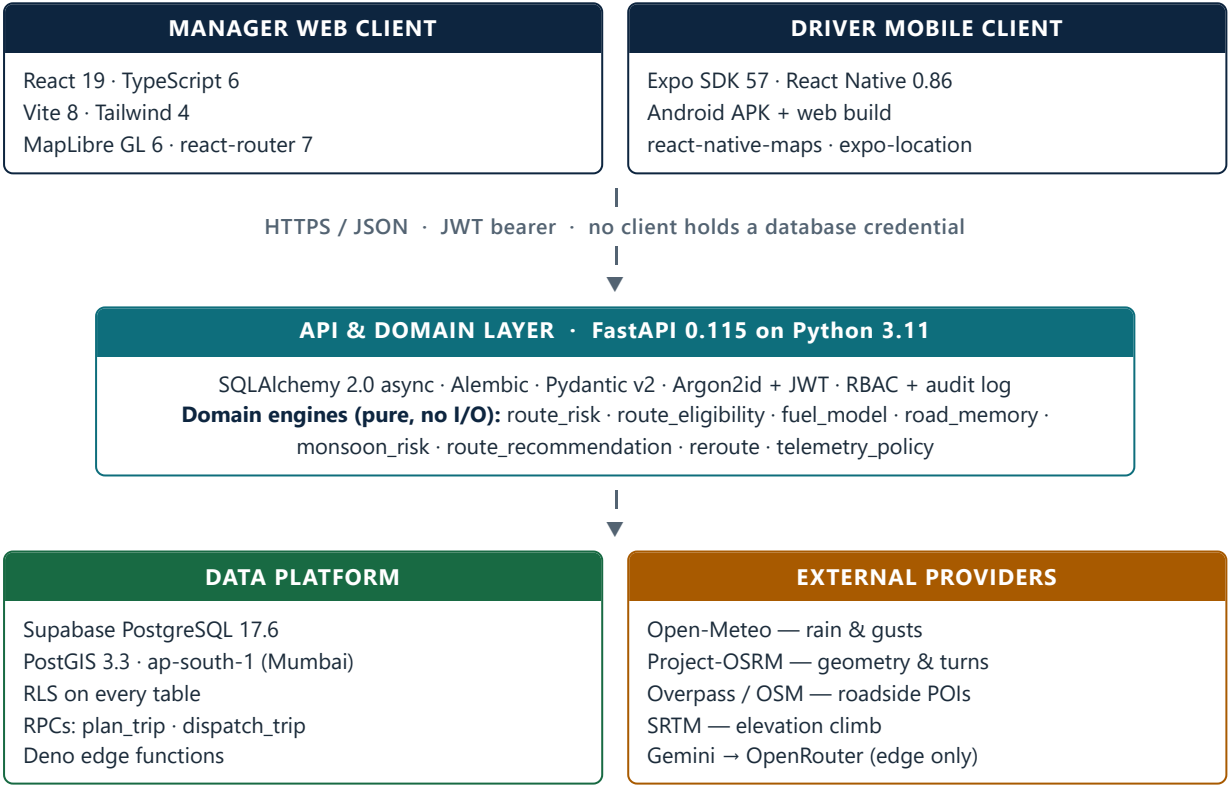
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Why each choice was made

1 Stack at a glance

Two purpose-built clients over one backend. Clients never hold a database credential and never talk to the data platform directly — which is what makes the deterministic safety rules impossible to bypass.



Layer	Core technology	Version pinned in the tree
Driver client	Expo SDK · React Native	expo ~57.0.18 · react-native 0.86.3 · react 19.2.3
Manager client	React · Vite · Tailwind	react ^19.2.8 · vite ^8.2.2 · tailwindcss ^4.3.3
API	FastAPI · Uvicorn	fastapi 0.115.6 · uvicorn[standard] 0.34.0
Language (backend)	Python	3.11.9 (>=3.11, <3.13 in pyproject)
Language (clients)	TypeScript	~6.0.2 (manager) · ~6.0.3 (driver)
ORM / migrations	SQLAlchemy · Alembic	sqlalchemy 2.0.36 · alembic 1.14.0
Database driver	psycopg 3 (binary)	psycopg[binary] 3.2.3
Geospatial ORM	GeoAlchemy2	geoalchemy2 0.16.0

Layer	Core technology	Version pinned in the tree
Database	PostgreSQL + PostGIS	PostgreSQL 17.6 · PostGIS 3.3 · region ap-south-1
Validation	Pydantic · pydantic-settings	pydantic 2.10.4 · pydantic-settings 2.7.0
Auth	Argon2id · PyJWT	argon2-cffi 23.1.0 · pyjwt[crypto] 2.10.1
Maps	MapLibre GL · Leaflet · RN Maps	maplibre-gl ^6.6.0 · leaflet ^1.9.4 · react-native-maps 1.27.2
Edge runtime	Deno (Supabase Edge Functions)	gemini-ai · driver-trip · resolve-map-link
Tests	pytest · Vitest	pytest 8.3.4 · vitest ^3.2.7 (mgr) / ^4.1.11 (drv)

2 Layer 1 — Driver mobile client

A phone-only field client. One hand, poor signal, sometimes at night, sometimes with no network at all. Every design constraint below follows from that.

Runtime & Framework		
Package	Version	Role in this project
expo	~57.0.18	Managed React Native toolchain; supplies the EAS build pipeline that produces the Android APK
react-native	0.86.3	Native runtime for Android
react	19.2.3	UI runtime, shared version line with the manager client
react-native-web	^0.21.2	Lets the same screens run in a browser for demo and for the Vitest suite
typescript	~6.0.3	Strict typing; <code>tsc --noEmit</code> is a release gate
expo-build-properties	~57.0.17	Android build configuration (SDK levels, native flags)

DEVICE CAPABILITIES

Package	Version	Why it is required
expo-location	~57.0.14	Foreground GPS streaming — the single input every live feature depends on
expo-secure-store	~57.0.2	Hardware-backed keystore for the refresh token; tokens never touch AsyncStorage
@react-native-async-storage/async-storage	2.2.0	Offline queue for GPS fixes and the cached route package
expo-speech	~57.0.2	Spoken turn guidance — a driver on a hill road should not read a screen
expo-clipboard	~57.0.1	Copy trip code / support reference without typing
expo-constants	~57.0.16	Reads the Expo host at runtime so the dev client finds the backend with no config
expo-status-bar	~57.0.1	Dark cockpit status bar treatment
react-native-safe-area-context	~5.7.0	Notch- and gesture-bar-safe layout for the map cockpit

MAPPING

Package	Version	Platform
react-native-maps	1.27.2	Android — native map surface, route polyline, live vehicle marker
leaflet	^1.9.4	Web build — same screen, browser-renderable, so the map is testable in CI
@types/leaflet	^1.9.22	Types for the web map path

The map component is split by platform extension — `DriverRouteMap.native.tsx` and `DriverRouteMap.web.tsx` — behind one import, so screen logic is written once and only the rendering surface differs.

Screens shipped in the driver app

Screen	What it does
LoginScreen	Email + password against the backend; token stored in SecureStore
AssignmentScreen	Truck verification — photo, registration, odometer, fuel level, damage notes
MapScreen	The cockpit: corridor polyline, next-turn banner, live speed, POI discovery, recentre / fit / mute controls
TripScreen	Stop list, arrive / complete actions, cargo detail, trip progress
SafetyScreen	Offline emergency guide, 112 / 108 / 1033 dial shortcuts, 45-minute rest tracker
AssistantScreen	Trip-scoped question answering in four sub-modes, backed by the edge AI function
TranslateBox	Machine translation of one short message, labelled as machine output
PhrasebookScreen	Reviewed offline phrase pack — works with no model, no server and no internet

Supporting modules

Module	Responsibility
src/tracking/tracker.ts	GPS acquisition, batching and de-duplication before upload
src/tracking/useLocationTracking.ts	Permission lifecycle, foreground subscription, revocation handling
src/auth/tokenStore.ts	SecureStore-backed token persistence and refresh rotation
src/api/client.ts	Backend selection (<code>EXPO_PUBLIC_BACKEND</code>), retries, error normalisation
src/trip/TripProvider.tsx	Server trip state synchronisation and offline-state transitions
scripts/check-release-config.mjs	Release gate — fails the EAS build if a LAN backend URL is configured, so a laptop-dependent APK cannot ship

3 Layer 2 — Manager web client

A desk tool: multiple screens, keyboard, dense information. Deliberately a separate application from the driver client because the two have genuinely different constraints, not because of platform fashion.

Package	Version	Role
react / react-dom	^19.2.8	UI runtime
react-router-dom	^7.18.3	Routing across the eight manager pages
vite	^8.2.2	Dev server and production bundler
@vitejs/plugin-react	^6.1.0	Fast refresh and JSX transform
tailwindcss + @tailwindcss/vite	^4.3.3	Utility styling; the GIS console theme is expressed as tokens
maplibre-gl	^6.6.0	Fleet map — vector tiles, planned corridor vs observed breadcrumb, worker-based rendering
@supabase/supabase-js	^2.116.0	Client for the Supabase-direct read path and RPC calls
typescript	~6.0.2	Strict typing; <code>tsc -b --noEmit</code> is a release gate
oxlint	^1.79.0	Linting (Rust-based, chosen for speed on a large surface)
vitest + @testing-library/*	^3.2.7	Component and page tests in a jsdom environment

Pages and components

PAGES

- `LoginPage` — email + password, role-aware landing
- `FleetPage` — live map, KPI bar, truck drawer
- `TripsPage` — atomic plan, dispatch, cancel, close
- `DriversPage` — roster, documents, deactivation
- `TrucksPage` — fleet register, capacity, retirement
- `AssignmentsPage` — driver↔truck pairing and verification
- `ReviewPage` — authorised route review queue
- `SystemPage` — backend health, provider status

KEY COMPONENTS

- `FleetMap` — MapLibre canvas, worker build verified by a test
- `FleetKpiBar` — Active / Moving / Stale / No-contact
- `TruckContextDrawer` — cargo utilisation, live speed, call driver
- `RouteRiskComparison` — the 3-corridor stepper with the fuel bar chart
- `TripRouteReview` — authorisation modal and review states
- `AddressPicker` — geocode search with a map-pick fallback
- `track.ts` — observed GPS breadcrumb rendering

4 Layer 3 — API and domain services

Package	Version	Role
<code>fastapi</code>	<code>0.115.6</code>	HTTP layer, dependency injection, OpenAPI docs at <code>/docs</code>
<code>uvicorn[standard]</code>	<code>0.34.0</code>	ASGI server
<code>pydantic</code>	<code>2.10.4</code>	Request/response contracts; every API shape is a typed model
<code>pydantic-settings</code>	<code>2.7.0</code>	Environment configuration with validation at startup
<code>sqlalchemy</code>	<code>2.0.36</code>	Async ORM, typed <code>Mapped[]</code> columns
<code>alembic</code>	<code>1.14.0</code>	Versioned schema migrations (9 revisions)
<code>psycopg[binary]</code>	<code>3.2.3</code>	One driver serving both the async app engine and Alembic's sync engine
<code>geoalchemy2</code>	<code>0.16.0</code>	PostGIS geometry/geography column types in the ORM
<code>httpx</code>	<code>0.28.1</code>	Async client for every external provider call, and the test transport

Application structure

backend/app/	
+-- main.py	app assembly, middleware, router mounting
+-- api/	HTTP surface
+-- auth.py	login · refresh · logout · me
+-- fleet.py	drivers · trucks · assignments
+-- trips.py	shipments · trips · routes · risk · reroute · fleet/active
+-- driver.py	the driver's own trip, GPS ingest, offline + nav packages
+-- geocoding.py	address suggest / details
+-- ai.py	AI availability and question answering
+-- health.py	/health (liveness) · /ready (dependency-checked)
'-- deps.py	auth + permission dependencies
+-- domain/	PURE LOGIC – no I/O, no DB, no provider
+-- route_risk.py	weighted risk rule, published constants
+-- route_eligibility.py	may this route be taken at all
+-- route_recommendation.py	explainable comparison + evidence parity
+-- reroute.py	NO_ACTION / ALERT_ONLY / PROPOSE
+-- fuel_model.py	CMEM-style physics fuel + CO2
+-- road_memory.py	5-state road evidence machine
+-- monsoon_risk.py	recurrence · rainfall · season · status
+-- route_progress.py	progress along a corridor from GPS
+-- trip_state.py	legal trip state transitions
+-- telemetry_policy.py	LIVE / STALE / NO_CONTACT thresholds
+-- weather.py	observation model + freshness
+-- landslide.py	exposure vs current hazard
+-- places.py	POI categories and ranking
+-- routing.py	provider-agnostic route model
'-- ai_prompts.py	prompt templates and refusal rules
+-- services/	orchestration: DB + providers + domain
+-- models/	SQLAlchemy tables and enums
+-- schemas/	Pydantic request/response contracts
+-- core/	config · security · permissions · rate limit · errors
+-- db/	engine, session, provider selection
'-- auth/	token verification

THE ARCHITECTURAL RULE THAT MAKES THIS TESTABLE

app/domain/ contains no imports of SQLAlchemy, httpx or FastAPI. Every engine is a pure function over dataclasses. That is why the risk, fuel and eligibility logic can be exercised exhaustively without a database, and why the constants can be argued with in review.

Complete endpoint surface

Method	Path	Purpose
GET	/health	Liveness — checks no dependency, always 200 when the process is up
GET	/ready	Readiness — real PostgreSQL + PostGIS version strings, 503 when unhealthy
POST	/api/auth/login	Argon2id verification, access + refresh issue, rate limited per IP and per identifier
POST	/api/auth/refresh	Rotating refresh with reuse detection (family revocation)
POST	/api/auth/logout	Revoke the presented refresh token
GET	/api/auth/me	Current principal, role and permissions

Method	Path	Purpose
GET / POST	/api/drivers	List / create drivers
GET / PATCH	/api/drivers/{driver_id}	Read / update one driver
POST	/api/drivers/{driver_id}/deactivate	Soft-deactivate, audited
GET / POST	/api/trucks	List / create trucks
GET / PATCH	/api/trucks/{truck_id}	Read / update one truck
POST	/api/trucks/{truck_id}/retire	Retire a truck, audited
GET / POST	/api/assignments	List / create driver↔truck assignments
GET	/api/assignments/{assignment_id}	Read one assignment
POST	/api/assignments/{id}/verify	Manager review of the driver's verification submission
POST	/api/assignments/{id}/end	End an active assignment
GET / POST	/api/shipments	List / create shipments with cargo items
GET / POST	/api/trips	List / create trips
POST	/api/trips/plan	Atomic shipment + trip + stops creation with capacity validation
GET	/api/trips/{trip_id}	Full trip detail with stops, routes and events
POST	/api/trips/{trip_id}/dispatch	Dispatch to the driver
POST	/api/trips/{trip_id}/cancel	Cancel with reason, audited
POST	/api/trips/{trip_id}/close	Close a delivered trip
GET	/api/trips/{trip_id}/routes	The candidate corridors for this trip
POST	/api/trips/{trip_id}/routes/recalculate	Re-run the routing provider and re-score
GET	/api/trips/{id}/routes/{rid}/risk	Full risk assessment with components, reason codes and named gaps
GET	/api/trips/{id}/routes/recommendation	Explainable comparison across candidates
GET / POST / DELETE	/api/trips/{id}/routes/{rid}/review-authorization	Issue, read and revoke a single-use 30-minute reviewer token
POST	/api/trips/{id}/routes/{rid}/select	Select a corridor (consumes the authorisation when one is required)
GET	/api/trips/{trip_id}/reroute	Assessment: NO_ACTION · ALERT_ONLY · PROPOSE
POST	/api/trips/{trip_id}/reroute/accept	Human acceptance of a proposed corridor change
GET	/api/trips/{trip_id}/track	Observed GPS breadcrumb for the trip
GET	/api/fleet/active	Live fleet snapshot with freshness classification
GET	/api/driver/me	The signed-in driver's own profile
GET	/api/driver/me/assignment	Current assignment and its verification state
POST	/api/driver/me/assignment/verify	Submit truck verification evidence

Method	Path	Purpose
GET	/api/driver/me/trip	The driver's current trip
GET	/api/driver/me/trip/navigation	Route geometry + turn maneuvers, bound to a route revision
GET	/api/driver/me/trip/offline-package	Everything needed to keep working with no network
GET	/api/driver/me/trip/places	Roadside POIs: near me · along route · in viewport
POST	/api/driver/me/trip/accept	Driver accepts the dispatched trip
POST	/api/driver/me/trip/start	Start the trip
POST	/api/driver/me/trip/stops/{id}/arrive	Arrive at a stop
POST	/api/driver/me/trip/stops/{id}/complete	Complete a stop
POST	/api/driver/me/trip/complete	Complete the trip
POST	/api/driver/me/location	GPS ingest — idempotent on (trip_id, device_fix_id)
GET	/api/geocoding/suggest	Address suggestions
GET	/api/geocoding/details	Resolve a suggestion to a coordinate
GET	/api/ai/status	Whether a model is reachable, and which provider answered
POST	/api/ai/ask	Trip-scoped question; driver-scoped, bounded, stateless

5 Layer 4 — Data platform

Component	Detail
Primary database	Supabase-managed PostgreSQL 17.6 , region <code>ap-south-1</code> (Mumbai) — data stays in India
Geospatial extension	PostGIS 3.3 — <code>geography(Point,4326)</code> and <code>geography(LineString,4326)</code>
Connection mode	Session pooler on port 5432 (not the transaction pooler on 6543 — it breaks psycopg prepared statements and is unsafe for Alembic DDL)
Offline fallback	WSL2 Ubuntu + PostgreSQL 18 + PostGIS 3.6 — opt-in only, never an automatic fallback
Row Level Security	Enabled on every table with no policies, because Supabase publishes <code>public</code> through its Data API
Server-side RPCs	<code>plan_trip</code> , <code>dispatch_trip</code> , <code>select_route</code> — operations that must be atomic across several writes
Edge Functions (Deno)	<code>gemini-ai</code> , <code>driver-trip</code> , <code>resolve-map-link</code>
Migrations	9 Alembic revisions, listed below; Alembic follows <code>DATABASE_PROVIDER</code> so it can never migrate a different database than the one being served

Migration history

Revision	What it introduced
0001_bootstrap_postgis	PostGIS extension and spatial groundwork

Revision	What it introduced
0002_core_domain	Users, drivers, trucks, documents, assignments, shipments, trips, stops, routes, events, GPS points
0003_refresh_tokens	Rotating refresh-token family table
0004_audit_actor_restrict	Audit actor FK made restrictive so history cannot be orphaned
0005_trip_event_location_null	Trip events may legitimately have no location
0006_current_assignment_unique	Partial unique index — one active assignment per driver
0007_route_review_authorizations	Single-use reviewer tokens with evidence digest and expiry
0008_trip_driver_acceptance	Explicit driver acceptance timestamps
0009_route_maneuvers	Turn-by-turn maneuver storage bound to a route revision

Tables and column counts

Table	Columns	Notes
users	9	Argon2id hash, role, activity flag
drivers	16	Licence, emergency contact, salary (permission-gated)
driver_documents	12	Type, expiry, verification actor and timestamp
trucks	18	Capacity, axles, dimensions, baseline mileage, odometer
truck_documents	12	RC, insurance, fitness, PUC, permits
truck_maintenance	10	Service / repair / breakdown / inspection history
driver_truck_assignments	15	Verification photo, reported registration / odometer / fuel, mismatch flag
shipments	15	PostGIS pickup and destination points, priority, status
cargo_items	10	Weight, quantity, hazardous and perishable flags
trips	21	Selected route, planned vs current ETA, delay minutes, lifecycle timestamps
trip_stops	14	Sequence, kind, geofence radius, planned vs actual arrival
trip_routes	18	LineString geometry, distance, duration, fuel, risk score and factors, maneuvers, provider id
trip_events	9	Append-only lifecycle log with optional location and JSON payload
gps_points	13	Point geography, speed, heading, accuracy, <code>device_fix_id</code> for idempotency, mock-location flag
refresh_tokens	11	Family id, replacement chain, revocation reason, user agent, IP
audit_logs	10	Actor, action, entity, before / after JSON, reason, IP
route_review_authorizations	18	Evidence digest and snapshot, policy and evidence versions, issue / consume / revoke trail

6 Layer 5 — Intelligence and AI

THE HONESTY RULE THIS WHOLE LAYER IS BUILT AROUND

Nothing in this section is a trained machine-learning model. Every engine is a **deterministic weighted rule with published constants**, carries a version name that says so, and has a test asserting that no `confidence`, `probability` or `model_version` field ever reaches the wire. The language model is a separate thing entirely: it rewords facts the engines have already produced, and has no write path to any decision.

6.1 Deterministic engines

Engine	Kind	Method
route_risk	Weighted rule	Rain ≤ 45 pts · wind ≤ 25 · duration ≤ 20 · distance ≤ 10, plus landslide points. Bands: LOW < 30 ≤ MODERATE < 60 ≤ HIGH. Thresholds: rain moderate 2.5 mm, heavy 7.5 mm; gust notable 40 km/h, severe 60 km/h. Stale observations are counted and reported, never scored.
route_eligibility	Refusal gate	Runs <i>before</i> ranking. A confirmed closure is REJECTED, not penalised — a penalty can be out-voted by other components, a refusal cannot. Unknown landslide evidence forces REQUIRES_REVIEW; unknown roadmap factors do not.
fuel_model	Physics (CMEM / SAE J1321)	$F = (B + K_{\text{payload}} \cdot M + K_{\text{speed}} \cdot \max(0, v - 50)) \cdot D + K_{\text{grade}} \cdot \max(0, \Delta H)$ with published coefficients: B = 0.22 L/km, K_payload = 0.0075 L/(t·km), K_grade = 0.00018 L/m, K_speed = 0.002. CO ₂ at 2.68 kg/L.
road_memory	Evidence state machine	Five states: VERIFIED_OPEN · REPORTED_INCIDENT · CLOSED · REPAIR_REPORTED · AWAITING_VERIFICATION. Silence is not evidence — no elapsed time may move a road toward VERIFIED_OPEN. Evidence carries a source; FLEET_TRAVERSAL is the strongest.
monsoon_risk	Weighted rule	Recurrence · rainfall · season · last observed road status. Passability is a separate three-valued field, not a score, so a closed segment cannot narrowly lose a comparison.
route_recommendation	Comparison rule	Deltas in points, minutes and kilometres — never as a percentage, because a risk score is an ordinal on this project's own scale. Refuses to compare when the two routes were scored from different available inputs.
reroute	Threshold + comparison	Three outcomes: NO_ACTION · ALERT_ONLY · PROPOSE. ALERT_ONLY exists because most of the North East is a single corridor and "this road has deteriorated and there is nothing better" is real information.
telemetry_policy	Freshness rule	LIVE / STALE / NO_CONTACT classification from fix age; drives the fleet KPI bar and the map marker states.
route_progress	Geometric	Progress along the selected corridor computed from the observed GPS track.

6.2 The five-stage accessibility pipeline

1 · ROUTE	2 · WEATHER	3 · LANDSLIDE	4 · FUEL	5 · SYNTHESIS
OSRM returns geometry, duration and maneuvers for each candidate corridor.	Open-Meteo sampled at points along the polyline; each reading carries a freshness stamp.	Exposure to known zones, distinguished from a current hazard report.	Physics model over distance, payload, climb and speed → litres and CO ₂ .	Weighted score, band, reason codes, component breakdown and named unavailable factors.

FACTOR AVAILABILITY, STATED HONESTLY

Evidence factor	Status	Note
Weather (rain, gusts)	IMPLEMENTED	Real provider, freshness-aware; stale readings never lower a score
Landslide exposure	IMPLEMENTED	is_known separates exposure from a current hazard
Fuel model	IMPLEMENTED	Availability-gated; returns null rather than guessing
Distance / duration exposure	IMPLEMENTED	Kept separate from conditions via condition_points
Field reports	PARTIAL	UNVERIFIED state exists in the model
Flood	NOT AVAILABLE	Declared unavailable in the response rather than scored as calm
Road quality	NOT AVAILABLE	Same — named in unavailable
Truck restrictions	NOT AVAILABLE	Same
Historical incidents	NOT AVAILABLE	Same
Elevation / terrain factor	NOT AVAILABLE	Elevation feeds the fuel model but is not yet a risk factor

6.3 Language model surfaces

Element	Detail
Primary model	Google Gemini Flash via the gemini-ai Supabase Edge Function; default gemini-3-flash-preview , configurable through GEMINI_MODEL
Fallback model	OpenRouter, default google/gemma-4-31b-it:free , 15-second budget
Key handling	GEMINI_API_KEY and OPENROUTER_API_KEY live server-side in the edge function. No client holds either.
Status model	One buildStatus() for both routes: OFFLINE · CONFIGURED · USABLE · QUOTA_EXHAUSTED · FALLBACK. available is true only for USABLE and FALLBACK.
Defences	Adversarial prompt-injection guards, chain-of-thought stripping (cleanAiText), per-minute rate limiting
Local alternative	Ollama path (llama3.2:3b) exists behind looks_local() , which refuses a non-loopback host unless deliberately overridden
Hard boundaries	Driver-scoped (every route takes CurrentDriver , no id parameter). One concurrent generation. Prompt-length and output-token caps. Stateless — no conversation is stored.
Forbidden outputs	Medical advice, medicine names or doses; a road-is-clear verdict; a promise that help is coming; turning an unavailable fact into a value. Pinned by tests.

6.4 Models specified but deliberately not trained

The project has a written model-governance policy: every model must beat a stated baseline on a held-out set or it does not deploy, and the baseline stays in the code permanently as the fallback. Four models are specified in full — features, target, baseline, evaluation metric and failure behaviour — and none is trained, because the data to validate them does not exist yet.

Model	Baseline it must beat	Why it is not trained
Fuel AI LightGBM / XGBoost / Ridge	Physics heuristic over distance, load ratio and ascent	No real NER truck fuel logs exist. A physics-informed synthetic generator is specified, and any metric from it must be reported as "on synthetic data" every time it appears.
Route Risk AI Calibrated logistic regression	Rule table: impassable within 5 km → 1.0; heavy rain in a landslide zone → 0.7; monsoon → 0.4; else 0.1	Not enough labelled disruptions will accumulate before the deadline. Evaluated on PR-AUC, not ROC-AUC, because disruptions are rare.
ETA correction Residual regression	<code>engine_eta × static_multiplier</code>	Predicts the <i>correction</i> to the routing engine's ETA, so a failed model degrades to <code>correction = 0</code> — exactly the engine's own estimate.
Plate verification PaddleOCR	Manual manager photo review — also the permanent fallback	<code>unreadable</code> is the expected common case, not an error. OCR never blocks a driver and never auto-rejects an assignment.

WHY THE LANDSLIDE MODEL IS BLOCKED — SPECIFICALLY

Not "not built yet" — **not buildable yet from the sources that exist**. GSI publishes hazard forecasts and a historical inventory, ASDMA publishes damage reports, IMD publishes weather. **No surveyed source publishes "this road is open again."** Reopening is decided by PWD / NHAI / BRO and communicated ad hoc, so nothing machine-readable carries it. That means there is no ground truth for "was this road passable on this date" — which is the label any landslide or route-risk model would have to be trained and validated against.

7 Layer 6 — External data providers

Provider	Endpoint	Used for	Failure behaviour
Open-Meteo	<code>api.open-meteo.com</code>	Rainfall intensity, wind gusts along the corridor	Request still succeeds; the factor is marked <code>NOT_AVAILABLE</code> with a reason code. A 503 here would hide distance and duration evidence that is still perfectly good.
Project-OSRM	<code>router.project-osrm.org</code>	Road geometry, duration, turn maneuvers	Configurable primary URL with this as fallback; 8-second timeout
Overpass / OSM	<code>overpass-api.de</code>	Roadside POIs — fuel, tyre repair, hospitals, rest stops	Snapshot service degrades to cached results
SRTM elevation	NASA elevation data	Cumulative climb metres for the fuel model	No elevation → fuel model returns baseline or null, never an imputed value
Google Gemini	via Supabase Edge Function	Vernacular assistant, safety rewording, translation	Quota exhausted → OpenRouter fallback → otherwise every AI surface renders its unavailable state

Provider	Endpoint	Used for	Failure behaviour
OpenRouter	openrouter.ai/api/v1	Backup generation engine	15-second budget, then unavailable

THE PROVIDER CONTRACT, IN ONE LINE

A provider failure is displayed as a provider failure. It is never converted into a number. `UNKNOWN` $\neq$ `SAFE` and `STALE` $\neq$ `LIVE` are enforced in the domain layer, not left to the UI.

8 Security stack

Control	Technology	Detail
Password hashing	argon2-cffi 23.1.0	Argon2id (RFC 9106)
Tokens	pyjwt[crypto] 2.10.1	15-minute access token, 30-day refresh; the <code>crypto</code> extra is present so an RS256/JWKS verifier can be added without a dependency change
Refresh rotation	Token families	Every refresh issues a new token and revokes the old one. Reuse of a revoked token revokes the whole family — the standard detection for a stolen refresh token.
Rate limiting	In-process limiter	Login: 20/min per IP and 10/min per identifier. Refresh: 60/min per IP. 60-second window.
Authorisation	Permission constants	Four roles (ADMIN · MANAGER · DRIVER · AUTHORISED_REVIEWER) against ~28 named permissions such as <code>route:review_authorize</code> , <code>driver:read_sensitive</code> , <code>trip:execute_own</code> , <code>location:submit_own</code>
Audit	<code>audit_logs</code> table	Append-only: actor, action, entity, before / after JSON, reason, IP
Route authorisation	<code>route_review_authorizations</code>	Single-use, 30-minute, cryptographically bound to an evidence digest and a policy version
Database isolation	PostgreSQL RLS	Enabled on every table with no policies; no client holds a database credential
Transport	TLS + <code>DB_REQUIRE_SSL</code>	SSL required to the database by default
Secret hygiene	Startup validation	The backend refuses to start with the placeholder <code>SECRET_KEY</code> when <code>APP_ENV</code> is not development. <code>VITE_*</code> and <code>EXPO_PUBLIC_*</code> are inlined into client bundles and are therefore treated as public — no secret may use those prefixes.
Provider misconfiguration	Refusal, not fallback	<code>DATABASE_PROVIDER=supabase</code> with a localhost URL is rejected at startup rather than silently serving a different database.
Test-database guard	<code>tests/db_target.py</code>	A <code>do_connect</code> listener on SQLAlchemy's <code>Engine</code> class with an exact-match allowlist of one target — a wrong database is never dialled, rather than merely discouraged.

Control	Technology	Detail
Release gate	<code>check-release-config.mjs</code>	Refuses to build the APK when a LAN backend URL is configured, so a laptop-dependent build cannot ship

9 Testing and quality stack

Suite	Framework	Result	Environment
Backend	pytest 8.3.4 + pytest-asyncio + httpx	959 passed , 5 skipped, 0 failed	Isolated PostgreSQL 17 + PostGIS cluster on <code>127.0.0.1:55432</code> , target-guarded
Manager web	Vitest 3.2.7 + Testing Library + jsdom	124 passed (12 files)	jsdom
Driver app	Vitest 4.1.11 + react-native-web + jsdom	490 passed (36 files)	jsdom over react-native-web
Type checking	tsc	Clean on both clients	<code>tsc -b --noEmit</code> / <code>tsc --noEmit</code>
Lint	oxlint 1.79	—	Manager web
Production build	Vite 8	Succeeds	Manager web bundle

The 5 backend skips are named and legitimate: one non-Windows event-loop test and four destructive migration tests gated behind `RUN_DESTRUCTIVE_MIGRATION_TESTS=1` , because they downgrade to base and drop every table.

NOTABLE TEST SUITES

Suite	What it pins down
<code>test_db_target_guard.py</code>	The run fails closed against any remote or unverified database, and refusals do not leak credentials
<code>test_route_recommendation.py</code>	Deterministic comparisons, evidence integrity, and that no confidence-like field reaches the wire
<code>test_fuel_model.py</code>	Unladen baseline, laden payload, elevation grade penalty
<code>test_gemini_ai.py</code>	Prompt-injection defence, medical-diagnosis refusal, rate limiter
<code>test_golden_path_e2e.py</code>	One complete manager + driver operational cycle end to end
<code>test_concurrency.py</code>	Behaviour under simultaneous writes
<code>test_schema_drift.py</code>	The ORM and the migration head agree
<code>MapScreen.test.tsx</code>	Position truthfulness, honest clock aging, permission revocation
<code>FleetMap.worker.build.test.ts</code>	The production build actually emits the MapLibre worker
<code>geminiAiHandler.test.ts</code>	Reasoning-token stripping and per-mode prompt specialisation

10 Build, release and deployment

Artifact	Toolchain and state
Android APK	EAS Build (hermetic). VersionCode 9 , build <code>3730b06b</code> , 71.29 MB , SHA-256 <code>228a1d3f5a7e3d2705fd818ebf6d7e275f84d74b61775133fbddf1d7ced437f9</code> . Passed a static security audit. Both <code>preview</code> and <code>production</code> profiles set <code>EXPO_PUBLIC_BACKEND=supabase</code> .

Artifact	Toolchain and state
Manager web bundle	Vite production build; host descriptors present for Netlify and Vercel (<i>only one should be kept — see below</i>)
Backend container	<code>backend/Dockerfile</code> and <code>backend/render.yaml</code> are written and ready
Edge functions	Deployed to the Supabase project; <code>gemini-ai</code> verified HTTP 200
Local orchestration	<code>docker-compose.yml</code> , <code>scripts/db-start.ps1</code> / <code>db-stop.ps1</code> for the optional offline database

OPEN DEPLOYMENT ITEMS, STATED PLAINLY

- **No hosted FastAPI deployment exists yet.** The container config is ready; a hosting account was not available on the build machine. Until it is deployed, manager *write* operations that are not covered by a Supabase RPC require the local backend.
- **Netlify and Vercel descriptors both exist** alongside `render.yaml` — three hosting descriptors for two deployables. One manager descriptor should be deleted before publishing so the app cannot be deployed twice with different environment variables.
- **`.runtime/` is protected by `.git/info/exclude`**, which is local-only and never committed. It should be moved into the tracked `.gitignore` before the repository is published.

11 Configuration surface

Variable	Default	Purpose
APP_ENV	development	Gates the placeholder-secret refusal
SECRET_KEY	– (required)	JWT signing key
ACCESS_TOKEN_EXPIRE_MINUTES	15	Access-token lifetime
REFRESH_TOKEN_EXPIRE_DAYS	30	Refresh-token lifetime
DATABASE_PROVIDER	supabase	<code>supabase</code> or <code>local</code> ; no automatic fallback between them
DATABASE_URL	–	Session-pooler connection string, rejected if it names a local host
LOCAL_DATABASE_URL	–	Offline-mode connection string
MIGRATION_DATABASE_URL	–	Optional separate DDL target
DB_POOL_SIZE / DB_MAX_OVERFLOW	5 / 10	Connection pool sizing
DB_CONNECT_TIMEOUT_SECONDS	5	Fail fast rather than hang a request
DB_REQUIRE_SSL	true	TLS to the database
ROUTING_ENABLED	true	Master switch for the routing provider
ROUTING_PRIMARY_URL	null	Self-hosted OSRM if available
ROUTING_FALLBACK_URL	<code>router.project-osrm.org</code>	Public OSRM
ROUTING_TIMEOUT_SECONDS	8.0	Routing budget
WEATHER_ENABLED	true	Master switch for weather sampling

Variable	Default	Purpose
WEATHER_PROVIDER_URL	api.open-meteo.com	Weather provider
WEATHER_TIMEOUT_SECONDS	6.0	Weather budget
GEOCODING_TIMEOUT_SECONDS	6.0	Address lookup budget
GOOGLE_PLACES_API_KEY	null	Optional richer place data
AI_ENABLED	true	Master switch for AI surfaces
GEMINI_API_KEY / GEMINI_MODEL	null / gemini-2.5-flash	Primary generation engine (edge function overrides to gemini-3-flash-preview)
GEMINI_RPM_LIMIT	10	Requests per minute ceiling
GEMINI_MAX_OUTPUT_TOKENS	512	Output cap
OLLAMA_BASE_URL / AI_MODEL	127.0.0.1:11434 / llama3.2:3b	Local model path
AI_ALLOW_NON_LOCAL_HOST	false	Structural guard — a driver's trip cannot be sent to a hosted endpoint by accident
AI_MAX_PROMPT_CHARS	4000	Prompt ceiling
AI_MAX_OUTPUT_TOKENS	200	Local-model output ceiling
RATE_LIMIT_ENABLED	true	Master switch for rate limiting
LOGIN_RATE_LIMIT_PER_IP	20	Per 60-second window
LOGIN_RATE_LIMIT_PER_IDENTIFIER	10	Per 60-second window
REFRESH_RATE_LIMIT_PER_IP	60	Per 60-second window
CORS_ORIGINS	—	Allowlist for the manager and Expo web origins
VITE_BACKEND / EXPO_PUBLIC_BACKEND	supabase	Client backend selection — public by construction

12 Why each choice was made

Choice	Reasoning
Two apps, one backend	The manager works at a desk with several screens; the driver works one-handed in a cab with poor signal, sometimes at night. Those are genuinely different design constraints, so one responsive web app would compromise both.
Clients never touch the database directly	Capacity rules, document enforcement and the safety path are only unbypassable if there is exactly one way to write. No client holds a database credential.
PostgreSQL + PostGIS	Corridor geometry, breadcrumb tracks and <code>ST_DWithin</code> stationary-vehicle detection are first-class database operations, not application loops.
Supabase in <code>ap-south-1</code>	Managed Postgres with PostGIS, edge functions and RLS, hosted in India — which matters for a government-sponsored logistics dataset.
psycopg 3, one driver	Serves both the async application engine and Alembic's sync engine from a single connection URL, so there is no second driver to keep in sync.
Python 3.11, not 3.14	LightGBM and PaddleOCR — the ML phase dependencies — do not yet publish 3.14 wheels. <code>pyproject.toml</code> pins <code>>=3.11, <3.13</code> deliberately.
Start with <code>run.py</code> , not <code>uvicorn</code>	On Windows the asyncio selector policy must be set before uvicorn creates its event loop, or every database call fails with a psycopg <code>InterfaceError</code> .

Choice	Reasoning
MapLibre + OpenStreetMap	Licence-free, self-hostable, and no per-map-load billing that would make a fleet console expensive to leave open all day.
Expo / React Native	One Android build from one TypeScript codebase, and <code>react-native-web</code> makes the same screens testable in CI without a device farm.
Gradient boosting (when models arrive)	Small tabular dataset with a non-linear gradient interaction. A tree model can also be explained to a judge in one sentence; a neural network cannot.
Deterministic engines instead of ML today	There is no ground truth for road passability, so a model could not be shown to beat a baseline. Shipping a published rule that can be argued with is the honest position, and the evidence log it produces is what a future model would train on.
No LLM in the safety path	After a real incident, "the truck had not moved more than 50 m in 60 minutes and the driver did not answer in 30" is defensible. "The model scored 0.83" is not.

DOCUMENT PROVENANCE

Compiled 9 September 2026 from the working tree: `backend/requirements.txt` , `backend/pyproject.toml` , `manager-web/package.json` , `driver-app/package.json` , `backend/app/**` , `backend/alembic/versions/**` , `supabase/functions/**` , and the reports in `docs/` . Test counts are from the runs recorded in `docs/SIH26002_GAP_MATRIX.md` (8 September 2026).